



SERIES OF EXERCISES N° 2: SEQUENCES OF REAL NUMBERS

Exercise 1.

Which of the following assertions are true? Justify correct ones and give a counter example for each incorrect assertion.

1. If a sequence (u_n) is strictly increasing and bounded above satisfying $\forall n \in \mathbb{N}, u_n \leq 2$ then it converges to 2.
2. If a sequence converges, then it is monotone and bounded.
3. If (u_n) is bounded, then it is convergent.
4. All sequences are either increasing or decreasing.
5. If (u_n) is convergent and (v_n) is divergent, then $(u_n + v_n)$ is divergent.
6. If (u_n) is convergent and (v_n) is divergent, then $(u_n \times v_n)$ is divergent.
7. If (u_n) and (v_n) are both divergent then so is $(u_n + v_n)$.
8. If (u_n) and (v_n) are both divergent, then so is $(u_n \times v_n)$.
9. If (u_n) converge to l , then (u_{n^2}) converge to l .
10. If a subsequences (u_{2n}) and (u_{2n+1}) are both convergent, then so is (u_n) .

Exercise 2.

By using (ε, N_0) definition of the limit, show that:

$$1. \lim_{n \rightarrow +\infty} \frac{\ln(1 + e^n)}{2n} = \frac{1}{2} \quad 2. \lim_{n \rightarrow +\infty} \frac{n^2 + 1}{n - 10} = +\infty$$

Exercise 3.

Determine the limits of the following sequences Find the limit if there exist, as $n \rightarrow +\infty$, of the following:

$$\begin{aligned} a_n &= \sqrt{n^4 + 3n - 1} - n^2, & b_n &= \left(1 + \frac{1}{n}\right)^n, & c_n &= \frac{\ln(n+e^n)}{n}, \\ d_n &= \frac{2n+(-1)^n}{5n+(-1)^{n+1}}, & e_n &= \frac{1+(-1)^n}{n}, & f_n &= \frac{\sin(n)+3\cos(n^2)}{\sqrt{n}}, \\ g_n &= \frac{(n+1)!+(n+2)!}{(n+3)!}, & h_n &= \frac{1}{n^2} \sum_{k=1}^n E(kx), & v_n &= \frac{E\left(\left(n+\frac{1}{2}\right)^2\right)}{E\left(\left(n-\frac{1}{2}\right)^2\right)}. \end{aligned}$$

Exercise 4.

Discuss the convergence or divergence of:

$$1. u_n = \sum_{k=0}^n \frac{n}{n^2 + k}, \quad 2. u_n = \frac{\lfloor nx \rfloor}{n^\alpha} \text{ avec } x, \alpha \in \mathbb{R}, \quad 3. u_n = \frac{\ln(n!)}{n}.$$

Exercise 5. Let $(u_n)_{n \geq 0}$ be a real sequence which is decreasing and satisfies $\lim_{n \rightarrow \infty} u_n = 0$. Define the sequence $(w_n)_{n \geq 0}$ by

$$w_n = \sum_{k=0}^n (-1)^k u_k \quad (n \geq 0).$$

1. Show that the subsequences $(w_{2n})_{n \geq 0}$ and $(w_{2n+1})_{n \geq 0}$ are adjacent (i.e. one is increasing, the other is decreasing, and their difference tends to 0).
2. What can you conclude about the sequence (w_n) ?
3. Deduce that the sequence with general term $s_n = \sum_{k=0}^n \frac{(-1)^k}{k+1}$ is convergent.

Exercise 6. Consider two real sequences $(u_n)_{n \geq 0}$ and $(v_n)_{n \geq 0}$ defined by

$$u_0 > 0, \quad v_0 > 0,$$

and, for all $n \in \mathbb{N}$,

$$u_{n+1} = \frac{u_n + v_n}{2}, \quad v_{n+1} = \frac{u_n + \sqrt{u_n v_n} + v_n}{3}.$$

Show that both sequences converge, that they have the same limit ℓ , and that this limit satisfies

$$v_1 \leq \ell \leq u_1.$$

Exercise 7. Consider the recursive sequence $(u_n)_{n \geq 0}$ defined by

$$0 < u_0 < 1, \quad u_{n+1} = \frac{u_n^2 + u_n}{2} \quad \text{for all } n \geq 0.$$

1. Show that for all $n \in \mathbb{N}$, $0 < u_n < 1$.
2. If the sequence (u_n) is convergent, determine the possible values of its limit.
3. Show that the sequence (u_n) is convergent.
4. Which of the possible limits is the actual limit of (u_n) ?

Exercise 8. 1. Let the sequence $(u_n)_{n \geq 2}$ be defined by

$$u_n = \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots + \frac{1}{n^2} = \sum_{k=2}^n \frac{1}{k^2}.$$

(a) Prove that (u_n) is a Cauchy sequence. Hint: Use the inequality $\frac{1}{n^2} \leq \frac{1}{n-1} - \frac{1}{n}$ ($n \geq 2$).

2. Define the sequences $(w_n)_{n \geq 1}$ and $(x_n)_{n \geq 2}$ by

$$w_n = \sum_{k=1}^n \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}, \quad x_n = \sum_{k=2}^n \frac{1}{\ln k} = \frac{1}{\ln 2} + \frac{1}{\ln 3} + \cdots + \frac{1}{\ln n}.$$

- (a) Show that (w_n) is not a Cauchy sequence.
- (b) Show that (x_n) is not a Cauchy sequence.
- (c) Show that both (w_n) and (x_n) diverge to $+\infty$.